

ECON 670 Project

Your project is done throughout the semester, and there are multiple places in which you will provide your work for grading. You are forecasting a single macroeconomic time series which you can download from a website. We will use the St. Louis Federal Reserve's FRED ("Federal Reserve Economic Data") database. <http://fred.stlouisfed.org>

Here are 6 alternative series for you to forecast. These are all prices of one kind or another. They are daily prices.

1. The exchange rate between the US dollar and the South Korean won. The series on FRED is DEXKOUS.
2. The exchange rate between the US dollar and the Brazilian real. The series on FRED is DEXBZUS.
3. The exchange rate between the US dollar and the Canadian dollar. The series on FRED is DEXCAUS.
4. The price of a barrel of Brent crude oil in Europe. The series on FRED is DCOILBRENTU.
5. The price of a gallon of propane gas in Mont Belvieu, Texas. The series on FRED is DPROPANEMBTX.
6. The average daily price of a dozen eggs in the U.S. in April. (Notice, this is not the price on April 30, it is the average for the month.) The series on FRED is APU0000708111.

Your project is to download your choice of JUST ONE of these six series and turn in, by March 31 2025 a forecast of the its value on **April 30, 2025**. For eggs, your forecast is for the next month's average price, based on all you know on March 31.

Your grade for this project is a combination of your score on the rubric (which you will know in advance of April 30) and an accuracy score. *The accuracy score is 25% of the project score and is scored as 25 minus the absolute percent error of the forecast (i.e., the absolute value of the percent error between your forecast and the actual value on April 30.)*

Your submission to the D2L site should include the following:

1. A copy of your CSV file showing the data you used in creating your forecast. If you used FRED, list any additional variables you used along with the forecasted series.
2. A report that includes all your code, your explanation of the code, and all output you used to create your forecast. You are encouraged to use R Markdown to do this, it will make everything easier for both you and me.
3. Show your training data separately from your testing data and report how you selected the model you chose.
4. Finally, using only Word or other text document, write a memo of no more than 500 words that explains your model as if I am a customer who wants to buy your forecast. Explain how it works as if I cannot (or will not) read your code.

Again, all this is due MARCH 31. Your grade will be preliminarily posted within a week, and then adjusted with your accuracy score after May 1.

Rubric for forecasting project

Element	Excellent (13-15)	Good (10-12)	Needs Improvement (7-9)	Poor (6 or less)
Data selection and analysis	Uses relevant, reliable, and diverse data sources, appropriately cleans and transforms data, and provides clear justifications for data choices.	Selects mostly appropriate data, performs basic data cleaning and tidying, and explains data selection rationale.	Uses limited data sources, lacks proper data cleaning, or does not adequately explain data choices.	Uses irrelevant or unreliable data, with significant data cleaning issues and no clear explanation.
Economic model application	Accurately applies appropriate economic models and theories, clearly explains model assumptions and limitations, and uses relevant statistical techniques.	Applies mostly appropriate models, explains key assumptions, and utilizes basic statistical methods.	Uses inappropriate or overly simplistic models, lacks sufficient explanation of model assumptions, or misinterprets statistical results.	Fails to apply relevant economic models, lacks understanding of model assumptions, and demonstrates significant errors in statistical analysis.
Forecast accuracy	Scored as explained in text, will be scored separately from rest of rubric.			
Explanation of results (both in report and memo)	Clearly interprets the forecast results in the context of economic theory, provides insightful analysis of key factors influencing the forecast, and discusses potential policy implications.	Provides a basic interpretation of the forecast results, identifies some key factors impacting the economy, and offers a general discussion of policy considerations.	Offers limited explanation of forecast results, lacks depth in analysis of key factors, and may not adequately connect findings to economic theory.	Fails to provide a clear explanation of forecast results, lacks economic analysis, and presents minimal discussion of policy implications.
Presentation quality (report only)	Presents findings in a well-organized, visually appealing manner with clear graphs, tables, and concise written communication.	Presents information in a mostly organized format with appropriate visuals, but may lack some clarity in presentation.	Presentation is disorganized, visuals are not effective, and written communication could be improved.	Presentation is poorly organized, lacks clarity, and contains significant errors in visuals and written communication.
Citations and referencing	Accurately cites all data sources and references used, following appropriate academic formatting guidelines.	Mostly cites relevant sources, with minor formatting inconsistencies.	Contains significant omissions in citations or inconsistencies in formatting.	Lacks proper citations and referencing for data and information used.